

The task is find the equation of best line that will separate the green points from red points.

We can see that when

- $w_1=1$
- $w_2=1$ and
- $w_0=-98$

we get a line that separates green points from red points

Q. Is this the only best possible line that separates the green and red points?

- No

Q. Now let's say that we add new feature to the existing datapoints will the existing line be able to separate this datapoints?

- No it won't work since it is 3D the same equation of line won't work

Q. If we cannot work with the line what should we do?

- since it a 3 dimensional plane we would need a plane to separate the data points

Class 1			Class 2		
Height x_0	Weight x_1	Width x_2	Height x_0	Weight x_1	Width x_2
33	27	0	63	77	0
13	6.5	0	67	97	0
43	39	0	85.7	60	0
22.6	16	0	93.8	90	0
13	35	0	91.4	105	0
33.8	7.3	0	73	114.8	0
46	13	0	41.7	66.7	0
10	19	0	50.3	89.5	0
22	30	0	79	88	0
24	43	0	56	104.5	0

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

$$1 x_1 + 1 x_2 + -98 = 0$$

$$x_1 + x_2 - 98 = 0 \quad \text{— line}$$

- ① Will I be able to separate these points using line — No
- ② What should we do — plane
- ③ What is the eqⁿ of plane?

Q. What would the equation of the plane be?

- $w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$

- and this is called as 3D hyperplane

(2D) : $w_1 x_1 + w_2 x_2 + w_0 = 0 \rightarrow$ general "line" equation.
A 2-D Hyperplane.

(3D) : $w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$.
↓
general "plane" equation
OR.
A 3-D Hyperplane.

(dD) : $w_1 x_1 + w_2 x_2 + \dots + w_d x_d + w_0 = 0$.
↓
general equation for a
d-dimensional hyperplane!

Vectors

So far we were storing numbers and to store them in python we can create a list, tuple, pandas series, or a numpy array

What is a Vector?

- As per physics vector is something which is a magnitude and direction both
- In terms of coding we can say that it is a collection of numbers which are stored in a particular format like list, etc

Vector representation

How do we represent a vector?

- if x was a vector then we represent it as x with a bar on top of it $\bar{x}$
- or you'll also find at some places that it is represented as an arrow on top $\vec{x}$

If we have a vector with d dimensions then how do we represent it?

- We can represent it as a column vector or as a row vector
- But in Machine learning or Mathematics in general we always take vectors represented in column format

How do we represent a vector that belongs to D dimensions mathematically?

- $x \in \mathbb{R}^d$
- In English it is read as x belongs to set of all real numbers in D dimensions

Linear Algebra - Vectors

Tuples, Numpy arrays, pandas Series, list

Q1) What is a vector?

- * Both Magnitude & direction (physics)
- * Collection of numbers

Q2) How do we represent a vector?

If vector $\vec{x}$ has d dimensions we can write

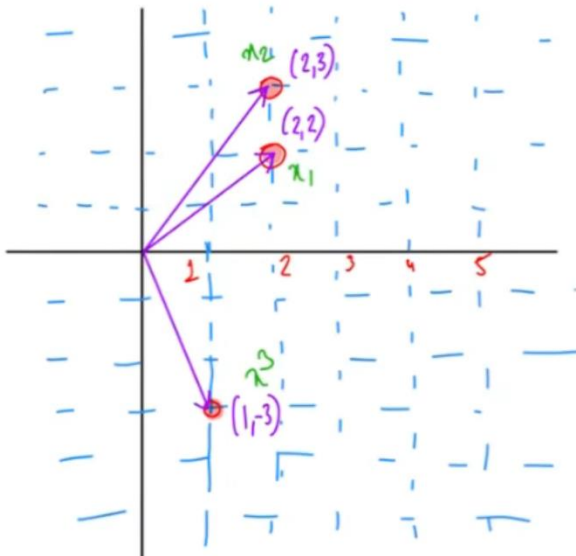
$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix} \text{ Column Vector}$$

$$\vec{x} = [x_1 \ x_2 \ x_3 \ x_4 \dots x_n] \text{ Row vector}$$

$\vec{x}, \vec{z}$ Set of all Real numbers in d dim
 $\mathbb{R}^d$
 $x \in \mathbb{R}^d$ belongs to

Visualizing a vector in 2D space

Linear Algebra - Vectors



$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\vec{x}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \vec{x}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \vec{x}_3 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Magnitude of vectors

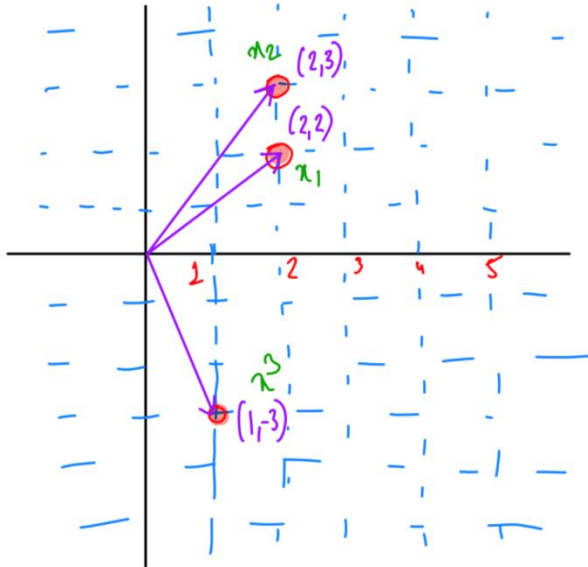
Q. We created a vector which starts from origin and ends at a point (x,y) now we want to find out what is the distance of the point from the origin.

- This is also called as the length or magnitude of the vector

Q. Now how do we compute this distance of a point from origin? can you recall any concept we studied in high school maths?

- We can simply apply pythagoras theorem to find this distance.
- thus the distance for 2d vector can be represented as

Linear Algebra - Vectors



$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\vec{x}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \vec{x}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \vec{x}_3 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Q) What are the magnitudes of these vectors?

It is the length of line joining point & origin

for 2D Vectors, magnitude is given by $\sqrt{x_1^2 + x_2^2}$

Q. Now how do we compute the same for a 3 dimensions Vector or let say a d dimension vector?

- We can generalize it as follows

- it is represented as $\|x\|$
- This is also known as **Euclidean Distance**

Then we have another type of L1 norm is?

- We call this a **Manhattan distance**

$\propto$ L_2 Norm: The formula for L_2 Norm of $\bar{x}$

$$\|\bar{x}\|_2 = \sqrt{x_1^2 + x_2^2 + x_3^2 \dots x_d^2}$$

similar to Euclidean distance

$\propto$ L_1 Norm: The formula for L_1 Norm of $\bar{x}$

$$\|\bar{x}\|_1 = |x_1| + |x_2| + |x_3| \dots |x_d|$$

similar to Manhattan distance.

Euclidean Distance

- It is the shortest distance between 2 points
- And it is defined as $\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}$

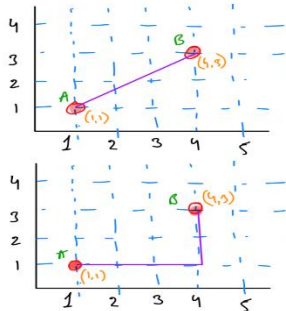
Manhattan Distance

It is nothing but the distance along the grid shaped roads on 2D plane

Different Types of Distances

① Euclidean Distance
Shortest Distance/path b/w A & B
 $\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2} = \sqrt{(3-1)^2 + (4-1)^2} = \sqrt{13}$

② Manhattan Distance
Distance by grid shaped roads on 2D plane.
 $|y_2 - y_1| + |x_2 - x_1| = |4-1| + |3-1| = 5$



- L_2 norm = Euclidean distance
- L_1 norm = Manhattan distance
- if no subscript is mentioned in $\|x\|$ then it means that we are using L_2 norm and is the default distance that we consider, until unless specified

Matrix Multiplication

I

Linear Algebra · Matrix Multiplication

$$A \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \quad B \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix}_{2 \times 2}$$

Dimension = # Rows $\times$ # Column.

* for Matrix Multiplication we need to check

$$A_{m \times n} \quad B_{a \times b} \rightarrow n = a$$

* Resultant Shape of matrix $\begin{bmatrix} \quad \end{bmatrix}_{m \times b}$

Q. Given two column vectors x and y of shape $2 \times 12 \times 1$ and $2 \times 12 \times 1$ can we multiply them?

- No we can't the reason is because as we saw that for matrix multiplication we need to check if $n=a$ if yes then only we can multiply them

Q. So now what can we do?

- We'll transpose to our rescue
- We can convert a column vector into a row vector and vice versa via the transpose operation
- if x^-x^- is a vector then it's transpose is represented as $x^-T x^-T$

Q. This operation of multiplying a transpose of vector x with column vector y is called as?

- This is known as **Dot product**

- And the notation of Dot product is $x^T \cdot y$ or $x^T y$
- To maintain this notation of dot product we always consider column vector and not row vectors

Linear Algebra. Angle b/w 2 Vectors.

$$\bar{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} \quad \bar{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2 \times 1}$$

if x is a vector
Transpose of Vector $\bar{x} = \bar{x}^T$

Transpose

$$\bar{x}^T = \begin{bmatrix} 1 & 2 \end{bmatrix}_{1 \times 2} \quad \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 1 \times 3 + 2 \times 4 \end{bmatrix}_{1 \times 1} = 11$$

This operation is called as DOT PRODUCT

$\bar{x}^T \bar{y}$ → Notation we use for dot product

Below is what will happen if we choose row vectors over column vectors

- You can see that we do not get a single scalar value as the output rather we are
Using loan example to understand dot product

Loan example

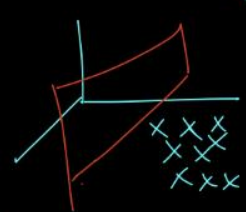
x_1 : applicant income "features"
 x_2 : Co-app income
 x_3 : credit history
 x_4 : prof area ..

Loan status "target"

$\underline{10} x_1 + \underline{4} x_2 + \underline{700} x_3 \rightarrow \text{Can this predict target}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $w_1 \quad w_2 \quad w_3$

$w = \begin{bmatrix} 10 \\ 4 \\ 700 \\ \vdots \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \end{bmatrix}$
 "new feature"
 $w^T x$

finding this
 is the biggest challenge
 "best decision boundary"



$\hookrightarrow$ depends on the parameters w & w_0
 $\hookrightarrow$ finding the "best" w & w_0
 is the main part of "learning"
 in machine learning

In the loan example, we must have done something like

$$10x_1 + 4x_2 + 700x_3 \rightarrow \text{target} \quad 10x_1 + 4x_2 + 700x_3 \rightarrow \text{target}$$

In this case, 10, 4 and 700 are the w_1 , w_2 and w_3 respectively, and their vector will be $[10, 4, 700]$

Taking this was w and $[x_1, x_2, x_3]$ as x

$$w^T x \quad w^T x = 10x_1 + 4x_2 + 700x_3 \quad 10x_1 + 4x_2 + 700x_3$$

- Our aim will be to find this w vector in such a way that we get the "best decision boundary"

9. Angle between two vectors

To calculate the angle between 2 vectors x and y we take $\cos(\theta)$

here x and y are both vectors and not individual coordinate points

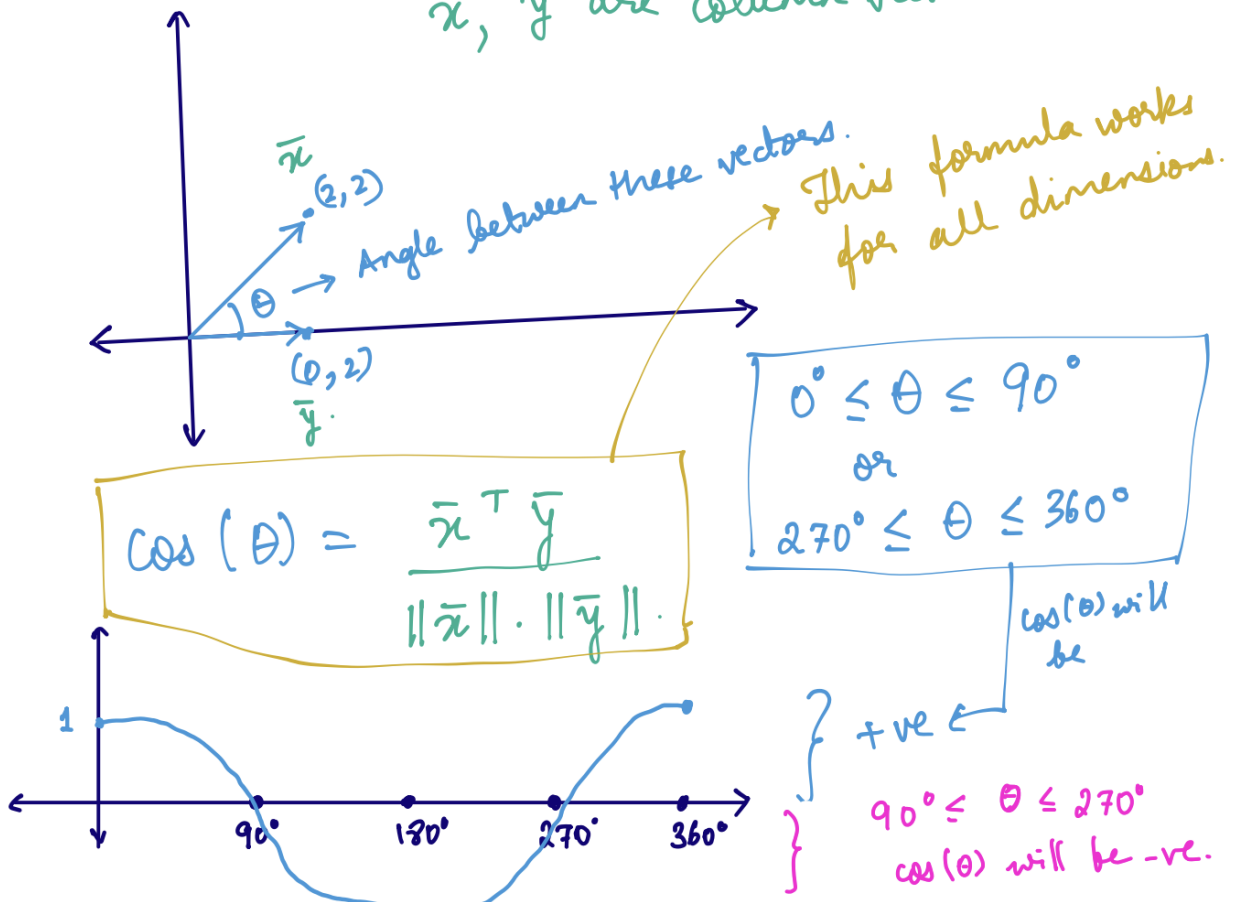
- θ is the angle between vector x and y

Now let's divide the 2d surface into 4 quadrants and based on the angle ie $\cos(\theta)$ we can conclude as follows

- If $\cos(\theta)$ value is greater than 0 then we can say that θ varies from 0 to 90 degrees or from 270 to 360 degrees.
- If $\cos(\theta)$ value is less than 0 then we can say that θ varies from 90 to 270 degrees.

Linear Algebra: Angle between two vectors.

$\bar{x}, \bar{y}$ are column vectors.



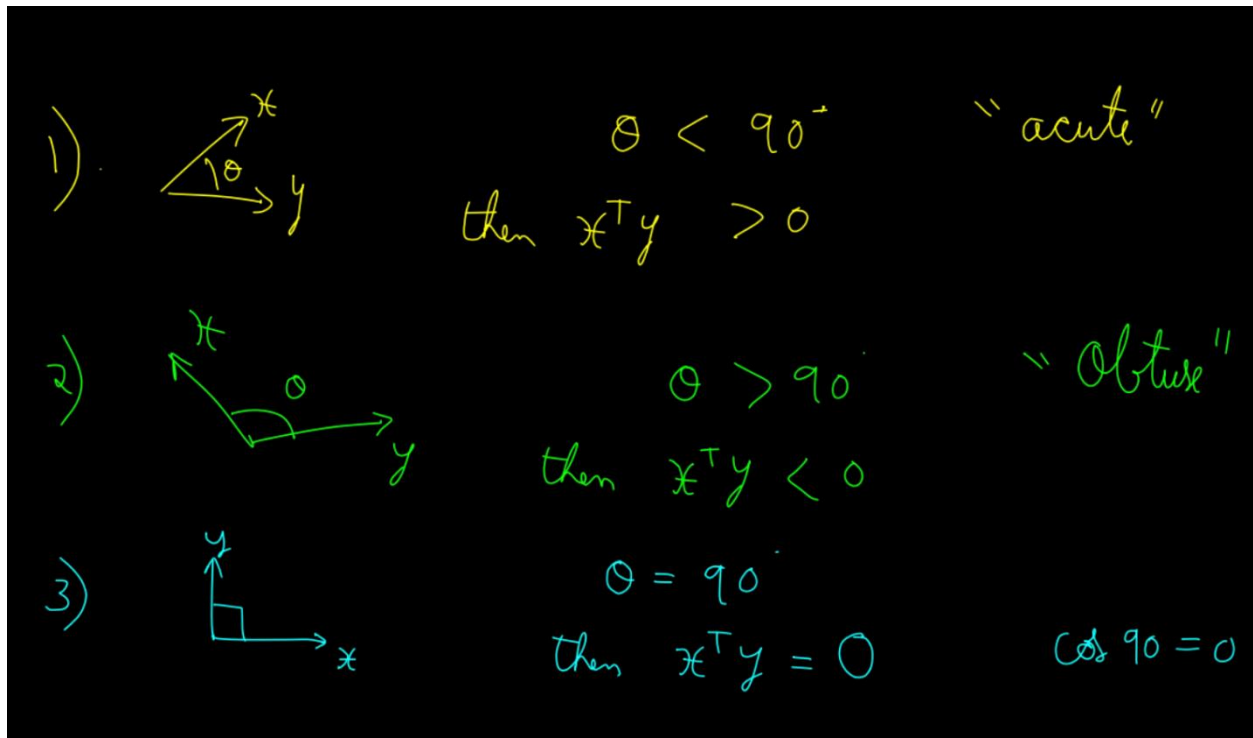
Q. Why do we care about the angle between 2 vectors?

- We'll learn in details why sign of dot product is important for now we can say that it'll help us understand where does a given vector lie ie in which quadrant with respect to other vector.

You can cover this briefly

Q. When we get a new point, how do we decide whether it lies in the positive or negative halfspace?

- If we look at the cosine graph, from 90-270 it lies in the negative y-axis.
- So based on the angles, we can find the position of the new point



12. Linking Coordinate Geometry and Linear Algebra

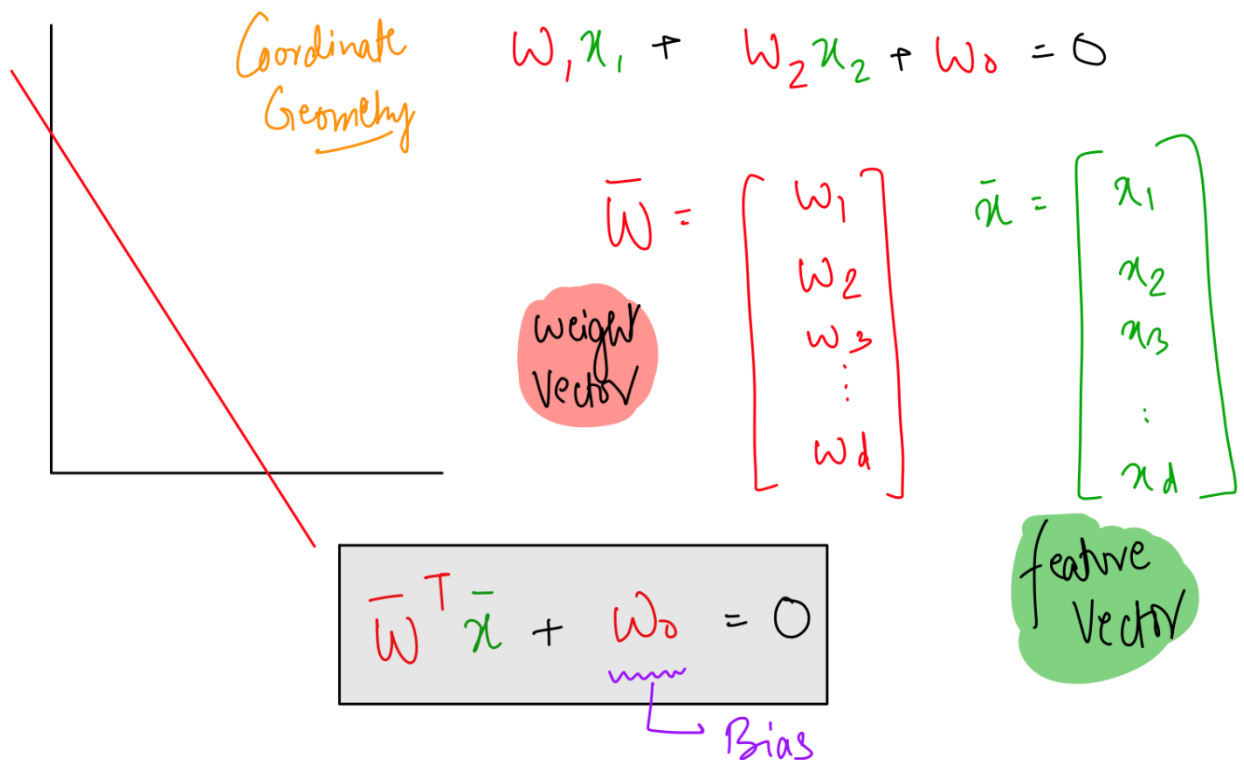
When we write $w_1x_1 + w_2x_2 = w_0 = 0$ this is in coordinate geometry but now if we have to write this equation for d dimensions then it will become a very long equation

Instead of this we create a neat hack which is as follows

- We store all the weight parameters together thus create a **weight vector** for us we denote this as capital W with a bar ie. $\bar{W}$
- Similarly we can store all the x 's as a single column vector and we denote this as lower case x ie $\bar{x}$
 - In Machine learning terms we call this as **Feature vector**

Notice that so far we have been using small case but since we are representing all the parameters as a single vector we denote it as a Upper case letter for the weight vector.

- $\bar{W}^T \bar{x} + w_0 = 0$



Now let say we have a equation of a line which is $w_1x_1 + w_2x_2 + w_0 = 0$

Q. For this above equation what is the weight vector?

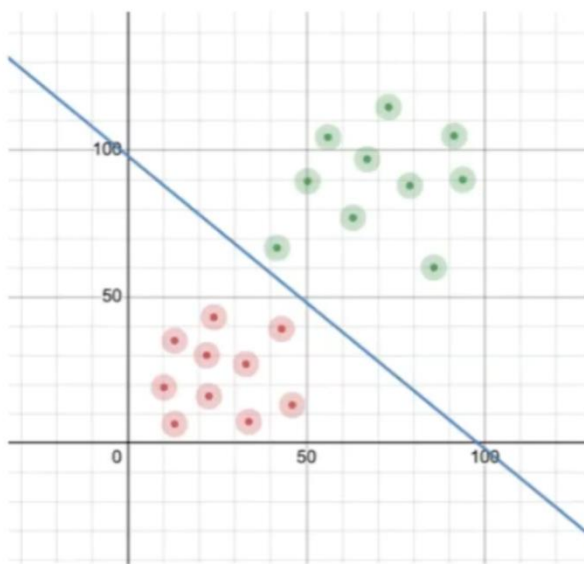
- $[1, 1]$ represented in column format

Q. The feature vector is?

- $[x_1, x_2]$ represented in column format

Q. Can we express this equation of line in terms of vector format that we saw above yes or no?

- Yes we can represent it as $\bar{w}^T \bar{x} + w_0 = 0$



$$x_1 + x_2 + w_0 = 0$$

$$\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

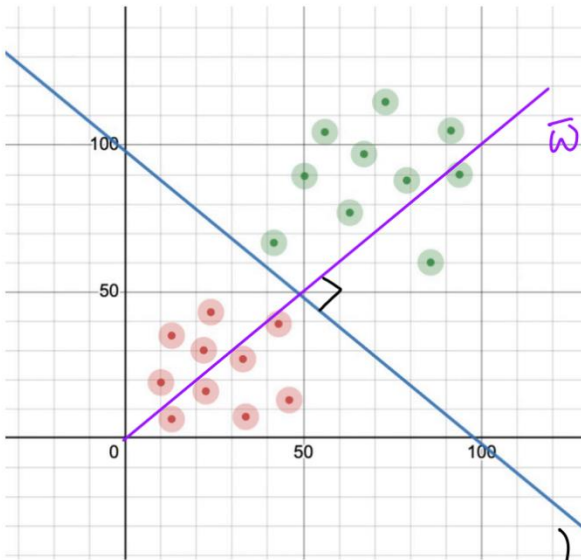
$$w_0 = -98$$

$$\bar{w}^T \bar{x} + w_0 = 0$$

$$x_1 + x_2 - 98 = 0$$

Now we can extend this line in magnitude keeping the direction same we can see that the weight vector is perpendicular to the equation of the line

- Thus point to remember is that the weight vector will always be perpendicular to the hyperplane



$$x_1 + x_2 - 98 = 0$$

$$x_1 + x_2 + w_0 = 0$$

$$\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$w_0 = -98$$

$$\bar{w}^T \bar{x} + w_0 = 0$$

Weight vector is $\perp$ to the hyperplane.

Q. Is the weight vector always going to be perpendicular to the hyperplane?

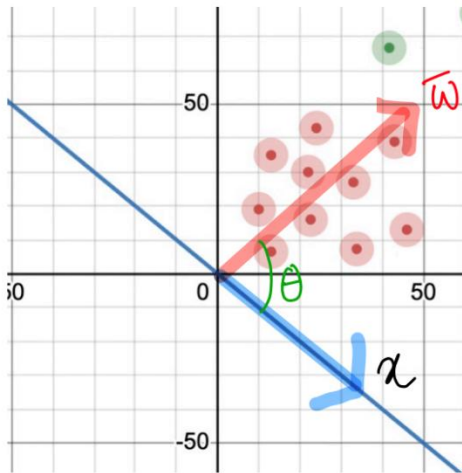
- The weight vector will always be perpendicular to the hyperplane, we'll see in the next class why it is always perpendicular.

Let's see this

Let's say that we have a equation of line which is $x_1 + x_2 = 0$ or $x_1 + x_2 = 0$

- here weight vector is $[1, 1]$ and w_0 is equal to 0
 - in vector notation terms we can represent it as $\bar{w}^T \bar{x} = 0$
- Let say we also have a vector x which is sitting on our line as show below and θ is the angle between the weight vector and vector x
 - to find the angle between 2 vector we use the $\cos(\theta)$ formula and we also know that $\bar{w}^T \bar{x} = 0$ if we replace this in the numerator we get $\cos(\theta) = 0$
 - We know that $\cos(\theta)$ is 0 when the angle is 90 or 270 degrees
 - Using this we have proved geometrically that weight vector is perpendicular to our line

In the next lecture we'll prove that weight vector is not only perpendicular to the line passing through origin it is perpendicular to any line we have



$$x_1 + x_2 = 0$$

$$\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\cos \theta = \frac{\bar{w}^T \bar{x}}{\|\bar{w}\| \|\bar{x}\|}$$

$$\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\bar{w}^T \bar{x} = 0$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

Unit vector

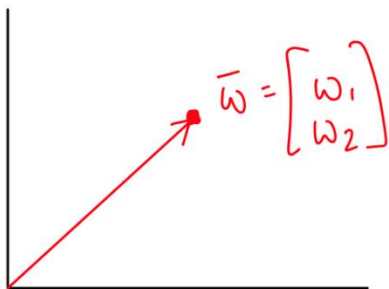
- These are vectors with direction but having magnitude as 1
- they help us find the direction of a vector

Unit Vector

"Unit Vector represents direction in which a vector is pointing towards."

Vector with magnitude = 1 $\Rightarrow$ Unit Vector

$$\|\bar{w}\| = \sqrt{w_1^2 + w_2^2}$$



$$\hat{\bar{w}} = \frac{\bar{w}}{\|\bar{w}\|}$$

$$\Rightarrow \hat{\bar{w}} = \begin{bmatrix} \frac{w_1}{\sqrt{w_1^2 + w_2^2}} \\ \frac{w_2}{\sqrt{w_1^2 + w_2^2}} \end{bmatrix}$$

$$\|\hat{\bar{w}}\| = \sqrt{\frac{w_1^2}{w_1^2 + w_2^2} + \frac{w_2^2}{w_1^2 + w_2^2}} = \sqrt{\frac{w_1^2 + w_2^2}{w_1^2 + w_2^2}} = 1$$

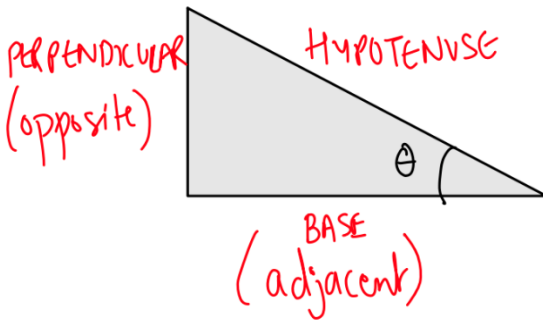
Basic Trigonometry

Instructor note:

- You can cover this briefly so that everyone is on the same page.

BASIC TRIGONOMETRY

$$\|\hat{w}\| = 1$$



$$\sin(\theta) = \text{perpendicular} / \text{hypotenuse}$$

$$\cos(\theta) = \text{base} / \text{hypotenuse}$$

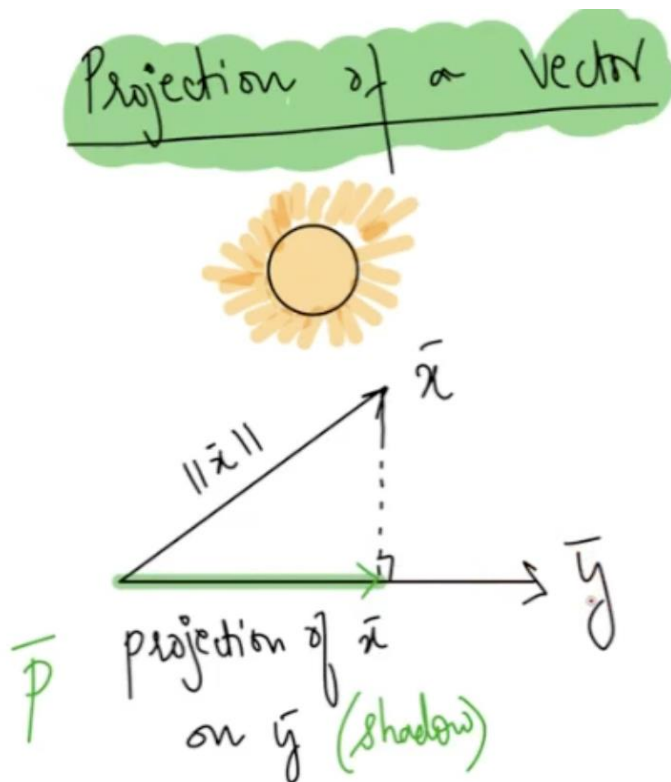
$$\tan(\theta) = \text{perpendicular} / \text{base}$$

Vector projection

Here we'll learn about projection of a vector onto another vector

we have two vectors x and y as shown and we want to find what is the length of the shadow that x casts on to the vector y when the sun is at the very top of the vector x

- To find this we draw a perpendicular line from the tip of the x vector and compute the length of the shadow that x casts onto y
- we call this shadow as **projection** and we say that we want to find the **length of projection of vector x on to y**
- let's denote this as another vector called as p now our task is to find the length of P vector in terms of x and y vectors



$$\bar{x} \cdot \bar{y} = \bar{x}^T \bar{y}$$

$$\cos \theta = \frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \|\bar{y}\|}$$

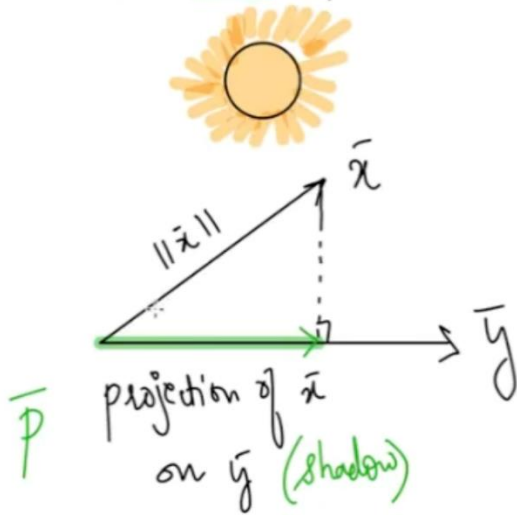
Q. Do you agree that the angle between vector x and y is the same as the angle the angle between vector x and vector P?

- Yes it'll be same because p lies in the same direction of vector y
- using this we can compute the angle between vector x and p

Also the angle between vector y and vector p would be 0 thus

getting $\cos[\theta] = 1 \cos(\theta) = 1$ we can re-arrange the terms to get the equation of norm of P vector

Projection of a Vector



$$\vec{x} \cdot \vec{y} = \vec{x}^T \vec{y}$$

$$\cos \theta = \frac{\vec{x}^T \vec{y}}{\|\vec{x}\| \|\vec{y}\|} \quad \text{b/w } \vec{x}, \vec{y}$$

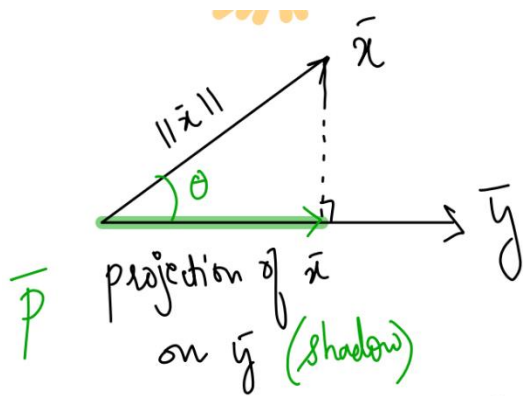
$$\underbrace{(\cos \theta)}_1 = \frac{\vec{p}^T \vec{y}}{\|\vec{p}\| \|\vec{y}\|} \Rightarrow \frac{\vec{p}^T \vec{y}}{\|\vec{y}\|} = \|\vec{p}\|$$

we know that $\cos(\theta)\cos(\theta)$ is base upon hypotenuse

- thus $\cos(\theta)\cos(\theta)$ between vector x and p would be $\|\vec{p}\| \|\vec{x}\| \|\vec{x}\| \|\vec{p}\|$

after solving for norm of p vector we'll get the length of projection of x vector onto y vector

- thus the norm of P vector is nothing but x vector in the direction of y



$$\cos \theta = \frac{\bar{p}^T \bar{y}}{\|\bar{p}\| \|\bar{y}\|} \Rightarrow \frac{\bar{p}^T \bar{y}}{\|\bar{y}\|} = \|\bar{p}\|$$

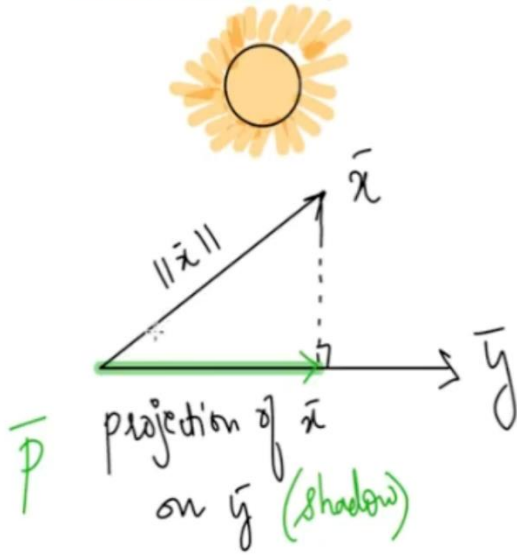
$$\cos \theta = \frac{\|\bar{p}\|}{\|\bar{x}\|}$$

$$\frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \|\bar{y}\|} = \frac{\|\bar{p}\|}{\|\bar{x}\|}$$

$$\|\bar{p}\| = \frac{\bar{x}^T \bar{y}}{\|\bar{y}\|} = \bar{x}^T \frac{\bar{y}}{\|\bar{y}\|}$$

$$\|\bar{p}\| = \bar{x}^T \hat{\bar{y}}$$

Projection of a Vector



$$\bar{x} \cdot \bar{y} = \bar{x}^T \bar{y}$$

$$\cos \theta = \frac{\bar{x}^T \bar{y}}{\|\bar{x}\| \|\bar{y}\|} \quad \text{b/w } \bar{x}, \bar{y}$$

$$\underbrace{\cos \theta}_1 = \frac{\bar{p}^T \bar{y}}{\|\bar{p}\| \|\bar{y}\|} \Rightarrow \frac{\bar{p}^T \bar{y}}{\|\bar{y}\|} = \|\bar{p}\|$$